MILITARY SPECIFICATION

CABLE ASSEMBLY - SWITCH, ELECTRICAL: M15

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of fuze option selector entitled Switch, Rotary: Fuze Option, Training, M15 which is used with the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8822418	- Switch, Rotary: Fuze Option, Training, M15.

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EL-8842339 - Equipment List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The rotary switch shall comply with all requirements specified on the drawing (dwg) 8822418, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Contact resistance. The electrical resistance measured between pin pairs of the cable plug shall not exceed 0.030 ohms when tested as specified in 4.4.1.

3.5 Insulation resistance. The insulation resistance between pins and between all pins and the case (except between those electrically connected) shall be not less than 20 megohms when tested as specified in 4.4.2.

3.6 Torque. The switch shall operate with a torque not exceeding 20 inch-pounds. The action of the switch shall not permit stopping between detent positions, it shall either return to the starting position or carry through to the opposite position. These conditions shall be met when the switch is tested according to 4.4.3.

3.7 Environmental testing. The rotary switch shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The rotary switch shall also comply with 3.4 and 3.5 before and after each environmental test. Testing shall be as specified in 4.4.4. Tests shall be conducted in the order specified.

3.7.1 Humidity. The fuze option switch shall withstand, without damage, moisture conditions as follows:

> Exposure to a temperature of 120 degrees Fahrenheit (°F) and 95 percent relative humidity for period of 15 days.

The test shall be as specified in 4.4.4.1.

3.7.2 Vibration. The rotary switch shall withstand vibration through the frequency range of 5 to 55 Hertz Hz at a double amplitude of 0.06 ± 0.006 inch for a period

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of thirty (30) minutes in each of three (3) major mutually perpendicular axes and after the test shall comply with the requirements of 3.4 and 3.5 when tested as specified in 4.4.4.2.

3.8 Soldering All soldering shall be in accordance with MIL-STD-1460.

3.9 Workmanship.

3.9.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.9.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.9.3 Burrs. Parts shall be free from burrs.

3.9.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.9.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.9.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 8 complete assemblies(see dwg 8822418) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

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4.2.2 Inspection and testing to be performed. The first article insection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I
First article inspection and tests

Inspection of Test	Dest	Non-Dest
1. Classification of Characteristics (see 4.3.2.1)		X
2. Contact resistance (see 3.4)		X
3. Insulation resistance (see 3.5)		X
4. For Humidity (see 3.7.1)	X	
5. Torque (see 3.6)		X
6. Vibration (see 3.7.2)	X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III),

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and inspected for all characteristics in the applicable classification. Any characterisics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristics inspections are performed.

Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Fuze selector, prior to potting (see dwg 8822418).

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
	101. Any component loose, damaged or missing	Visual/Manual
	102. Solder loose or insufficient (see 3.8)	Visual/Manual

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103.	Insulation inadequate or damaged	Visual
104.	Objective evidence relating to material missing or improper (see 3.1)	Visual

Minor:

201.	Evidence of poor workmanship (see 3.9)	Visual

4.3.2.1.2 Fuze selector (see dwg 8822418).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Location of tie points	Gage
102.	Length from switch to connector, min	Gage
103.	Hardware loose, damaged or missing	Visual/Manual
104.	Connector pins loose, damaged or missing	Visual/Manual
105.	Any component loose, damaged or missing	Visual/Manual
106.	Wires inadequately tied	Visual/Manual
107.	Pins in connector damaged, loose or missing	Visual/Manual
108.	Any operation incomplete or missing	Visual
109.	Potting inadequate	Visual
110.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.9)	Visual
202.	Marking improper, illegible or missing	Visual

4.3.2.1.3 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:

101. Preservation packing and packaging not in accordance with MIL-STD-1169.....Visual

102. Marking not in accordance with MIL-STD-1169.....Visual

Minor:

201. Evidence of poor workmanship (see 3.9).....Visual

4.3.3 Testing. All tests shall be performed. Tests shall performed in the order specified below:

Test	Test classification
a. Contact resistance	Major
b. Insulation resistance	Major
c. Torque	Major
d. Vibration	Major

Of these tests, d is considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Contact resistance, insulation resistance and torque. Each switch assembly in the lot shall be tested in accordance with 4.4.1, 4.4.2 and 4.4.3. If any item fails to comply with 3.4, 3.5 and 3.6, the item shall be rejected and removed from the lot.

4.3.3.2 Vibration. One sample rotary switch shall be selected at random out of every 25 rotary switches or portion thereof procuded. The sample shall be tested in accordance with 4.4.4.2. If any item fails to comply with 3.7.2, the lot shall be rejected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8842339). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).

b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

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4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Contact resistance. The electrical resistance measured between the following pin pairs of the cable plug shall not exceed the specified resistance when measured at the terminals of the assembly with a resistance bridge:

a. Full clockwise rotation of switch (switch viewed from shaft end).

A to B	H to J
B to H	K to L

b. Full counterclockwise rotation of switch (switch viewed from shaft end).

B to C

4.4.2 Insulation resistance. The fuze option switch shall be subjected to a direct current (dc) potential of 500 ± 50 volts until a steady static reading is obtained using a megohm tester from which a direct reading of resistance in megohms can be made.

4.4.3 Torque. Mount the switch on a rigid fixture. Measure the torque required to change positions of the switch in both the CW and CCW direction. Operate the switch slowly several times, observe action for possible "dead positions", ie., positions on which the switch might stop other than its two normal positions.

4.4.4 Environmental testing (see 3.7).

4.4.4.1 Humidity. The fuze option switch shall be tested in accordance with MIL-E-5272, Section 4.4.3, Procedure III. Within 2 hours after removal from the humidity chamber, the assembly shall be tested as specified in 4.4.1 and 4.4.2.

4.4.4.2 Vibration. The rotary switch shall be mounted to a rigid fixture. The fixture shall be mounted to a suitable vibration-producing apparatus.

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The rotary switch shall than be vibrated through a range of frequencies from 5 to 55 Hz at a constant double amplitude of 0.06 ± 0.006 inch for a duration of 30 minutes in each of its three (3) major mutually perpendicular axes. Each cycle, 5 to 55 Hz shall have a duration of one (1) minute and the frequency shall be varied uniformly throughout the cycle. The rotary switch shall show no damage after the vibration test and shall comply with the requirements of 3.4 and 3.5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The rotary switch shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The rotary switch shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The rotary switch shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

6.4 Supersession data. This specification supersedes the requirements of Purchase description PA-PD-1805 Amendment 1 dated 4 November 1964.

Custodian:
Army - PA

Preparing activity:
Army - PA

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